

# JULIAN TANUI

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[Portfolio](#)

## PROFESSIONAL SUMMARY

Software and AI Engineer with a BSc in Informatics and Computer Science, currently building production software for the climate technology sector at Verst Carbon. My work spans full-stack web applications, AI agent systems, and data pipelines, from a financial risk engine for carbon-credit portfolios to a real-time meeting intelligence platform and a deepfake detection system. Recognized for academic excellence through the Dean's List and experienced in international innovation challenges, I focus on turning complex problems into reliable, well-crafted products.

## TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Kotlin, SQL.
- **Frameworks:** Next.js, React, FastAPI, Flask, Node.js/Express, PyTorch, TensorFlow.
- **AI & Agents:** LLM agents (LangGraph, LangChain, multi-agent orchestration), model-selection routing across Claude/GPT-4o/Gemini, prompt engineering & evaluation frameworks.
- **Retrieval & ML:** RAG (FAISS, BM25, pgvector, ontology-backed retrieval), vector stores & semantic search, Computer Vision (CNNs, ResNet50), NLP.
- **Data Engineering:** ETL/ELT pipelines, data-quality/validation & freshness monitoring, geospatial sources (NASA FIRMS, GFW, GDELT, OpenStreetMap), interactive map dashboards.
- **Cloud & DevOps:** Docker, CI/CD, PostgreSQL/Supabase, Redis, MinIO, Modal, Inngest, Railway, Cloudflare Pages.
- **Tools:** Git/GitHub, Linux Systems, Cisco Networking, IoT Sensors.

## EDUCATION

**Strathmore University, Nairobi | Bachelor of Informatics and Computer Science**

*Coursework completed, December 2025 | Degree conferment: August 2026*

- **Honors:** Dean's List Award (2024-2025, 2025-2026) – Recognized for outstanding academic performance.
- **Leadership:** Local Committee Vice President for Finance and Legalities; Project Manager – SDG Hub, Strathmore.
- **Relevant Coursework:** Data Structures and Algorithms, Software Engineering, Machine Learning, Advanced Database Systems, Advanced Networking, Project Management.

## PROFESSIONAL EXPERIENCE

**Verst Carbon | SWE/AI Engineering Intern | Feb 2026 – Present**

- Designed and built multi-agent AI systems (LangGraph, LangChain) that automate multi-step analysis across geospatial, infrastructure, economic, and monitoring data, routing each task to the most suitable language model for accuracy and cost efficiency.
- Applied retrieval-augmented generation to ground AI responses in trusted sources, keeping outputs accurate and traceable.
- Built full-stack web applications with Next.js, React, and Python (FastAPI, Node.js), including authenticated dashboards, interactive maps, and real-time chat.

- Developed data pipelines that ingest and validate large geospatial and economic datasets, with automated quality and freshness checks.
- Containerized and deployed services with Docker and CI/CD, adding monitoring and evaluation to track reliability.

#### **CIC Insurance Group | ICT Intern | Jan 2025 – April 2025**

- Provided enterprise-level technical support, resolving complex ticket escalations and ensuring operational continuity.
- Managed network infrastructure, troubleshooting outages and optimizing performance for domain connectivity.
- Configured hardware assets (laptops, mobile devices), ensuring strict adherence to security protocols.

#### **Kenya National Library Service (KNLS) | Volunteer & Trainer | Jan 2024 – Mar 2024**

- Led digital literacy workshops, teaching computer proficiency to students and community members.
- Assisted in organizing educational workshops and fostering a collaborative learning environment.
- Assisted in front desk services and library resource organization.

### **HACKATHONS & INNOVATION CHALLENGES**

- **International Energy & Sustainability Hackathon (2026):** Completed the 2026 cohort, developing sustainable energy solutions.
- **Inflection AI Hackathon (Climate Change):** Developed 'EcoCoin,' a fintech solution for trading carbon credits for real currency to incentivize climate action.
- **Oracle Hackathon (Climate Change):** Built 'Smart IoT Farm,' an integrated system using sensors and AI to monitor soil health and automate irrigation.

### **SELECTED ENGINEERING PROJECTS**

#### **AI & Machine Learning**

- **Meeting Intelligence Engine (AI + Real-time Systems):** An end-to-end platform that captures meetings, produces live transcripts with speaker labels, and generates AI summaries, action items, and searchable chat. Includes calendar integration, secure access controls, and auto-generated weekly audio recaps. Built with GPT-4o on a React and Python stack.
- **DeepDetect (AI/ML):** An explainable deep-learning system for detecting deepfake images and video, trained on a large public dataset with strong accuracy. Provides visual explanations of its predictions, video analysis, and admin reporting. Built with a ResNet50 model, React/TypeScript, and Flask.
- **Project Rainmaker (AI Sales Agent):** An AI sales agent (LangGraph) that researches businesses online, evaluates their websites for marketing gaps, scores leads, and generates outreach reports. Developed for Okara AI's automated marketing platform.
- **Gaming RAG Assistant (AI):** A question-answering assistant over game documentation that combines semantic and keyword search for accurate, character-styled responses. Built with HuggingFace models and deployed on Streamlit Cloud.

#### **Climate & Fintech**

- **OfftakeOS (Fintech + AI Automation):** An automated financial risk engine for carbon-credit portfolios that ingests forward contracts, monitors real-world environmental and market data, and scores delivery risk on interactive dashboards. Built with Next.js, Flask, and PostgreSQL, backed by a comprehensive automated test suite.

- **CarbonVerify (Fullstack + DevOps):** A carbon-credit auditing dashboard with project mapping, sensor-data verification, and a certificate-issuing workflow. Built with React, a Node.js/Express API, PostgreSQL, and a full CI/CD pipeline.

#### **Mobile & Web**

- **CasaControl (Mobile):** A smart-home management mobile app built with Kotlin, enabling remote appliance control.
- **CommunityPlus (Web):** Web-based donation platform bridging donors and receivers to facilitate community support.

#### **ACHIEVEMENTS & CERTIFICATIONS**

- **Dean's List Award (2024-2025, 2025-2026):** Awarded for maintaining a high GPA and demonstrating academic leadership.
- **NSK AI RAG Bootcamp:** Completed intensive training on building Retrieval-Augmented Generation systems.
- **Cisco Certified:** IT Essentials (Cisco Networking Academy) – proficient in hardware assembly, maintenance, and security.
- **Cisco Certified:** Computer Networks (Cisco Networking Academy).

#### **LEADERSHIP & ACTIVITIES**

- **Local Committee Vice President** – Finance & Legalities, Strathmore University.
- **Project Manager** – SDG Hub, Strathmore University.
- **Member:** Google Student Developer Club (GDSC), Strathmore University.
- **Member:** Strathmore Computing and Engineering Sciences Association.
- **Registered Member:** The Kenya Red Cross Society.